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MATHEMATICS

0580/43

Paper 4 Calculator (Extended)

May/June 2026
2 hours

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You should use a scientific calculator where appropriate.
- You may use tracing paper.
- You must show all necessary working clearly.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For π , use either your calculator value or 3.142.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

 This document has **20** pages.

List of formulas

Area, A , of triangle, base b , height h .

$$A = \frac{1}{2}bh$$

Area, A , of circle of radius r .

$$A = \pi r^2$$

Circumference, C , of circle of radius r .

$$C = 2\pi r$$

Curved surface area, A , of cylinder of radius r , height h .

$$A = 2\pi rh$$

Curved surface area, A , of cone of radius r , sloping edge l .

$$A = \pi rl$$

Surface area, A , of sphere of radius r .

$$A = 4\pi r^2$$

Volume, V , of prism, cross-sectional area A , length l .

$$V = Al$$

Volume, V , of pyramid, base area A , height h .

$$V = \frac{1}{3}Ah$$

Volume, V , of cylinder of radius r , height h .

$$V = \pi r^2 h$$

Volume, V , of cone of radius r , height h .

$$V = \frac{1}{3}\pi r^2 h$$

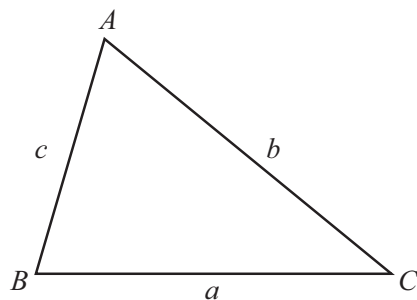
Volume, V , of sphere of radius r .

$$V = \frac{4}{3}\pi r^3$$

For the equation $ax^2 + bx + c = 0$, where $a \neq 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For the triangle shown,



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}ab \sin C$$





3

- 1 Complete the statement.

$$\frac{4}{7} = \frac{\boxed{}}{56}$$

[1]

- 2 Work out 55% of 12.2 .

..... [1]

- 3 Calculate $\frac{0.7 - 3.5^3}{4.2^2}$ giving your answer correct to 2 significant figures.

..... [2]





- 4 Kit and Jo are changing some money.
The exchange rate is \$1 = 0.93 euros.

(a) Kit changes \$350 into euros.

Calculate the amount, in euros, that Kit receives.

..... euros [1]

(b) Jo changes 150 euros into dollars (\$).

Calculate the amount, in dollars, that Jo receives.

\$ [1]

- 5 Ruta and Zoli share some money in the ratio 7 : 3.
Ruta receives \$420 more than Zoli.

Work out how much money Ruta and Zoli each receive.

Ruta \$

Zoli \$

[3]

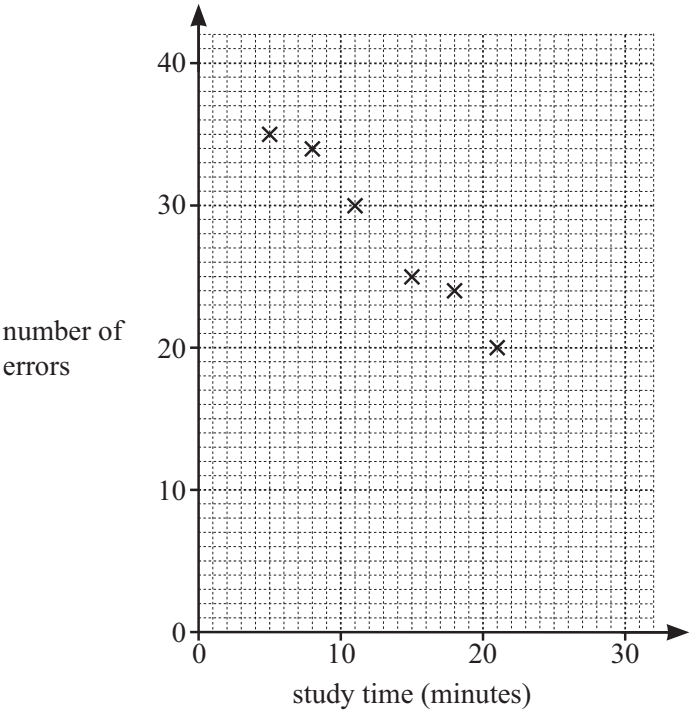


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6 The table shows the time 9 students each spent studying for a test and the number of errors each student made in the test.

study time (minutes)	5	8	11	15	18	21	25	27	31
number of errors	35	34	30	25	24	20	13	9	6

(a) Complete the scatter diagram.
The first 6 points have been plotted for you.



[2]

(b) What type of correlation is shown in the scatter diagram?

..... [1]

(c) Draw a line of best fit on the scatter diagram.

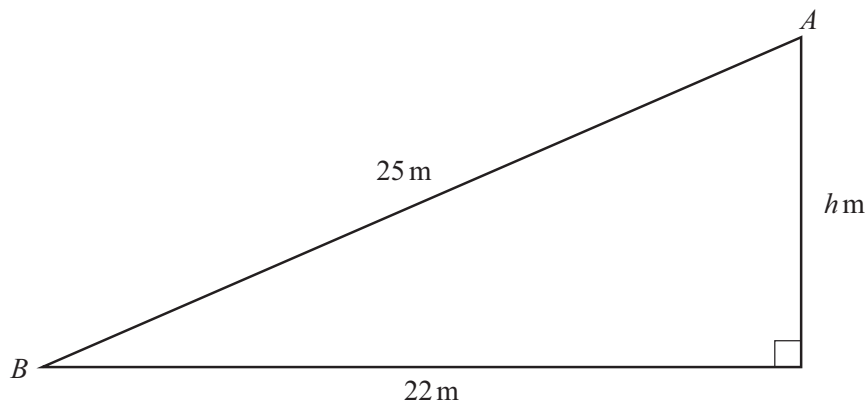
[1]

(d) Another student studies for 13 minutes.

Use your line of best fit to estimate the number of errors this student is expected to make in the test.

..... [1]





NOT TO SCALE

The point A is on the top of a vertical building.
The horizontal distance from the base of the building to B is 22 m.
 $AB = 25$ m and the height of the building is h m.

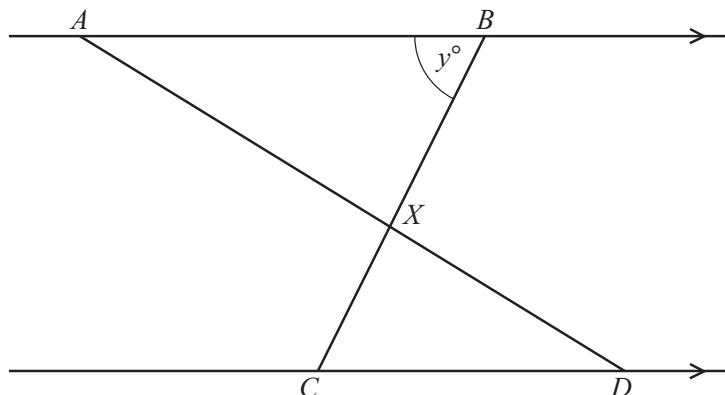
(a) Find the value of h .

$h = \dots\dots\dots$ [3]

(b) Work out the angle of depression of B from A .

$\dots\dots\dots$ [2]





NOT TO SCALE

AB and CD are parallel lines.
 AD and BC are straight lines that intersect at X .
 Angle $ABX = y^\circ$.

- (a) Complete the statement with a geometrical reason.

Angle $DCX = y^\circ$ because [1]

- (b) Triangle ABX is similar to triangle DCX .

$CX = 5$ cm, $XD = 8$ cm and $AD = 20$ cm.

- (i) Find BX .

$BX = \dots\dots\dots$ cm [2]

- (ii) Angle $CXD = 80.5^\circ$.

Work out the area of triangle CXD .

$\dots\dots\dots$ cm² [2]



- 9 The table shows the marks of 100 students in a test.

mark	$0 < x \leq 20$	$20 < x \leq 40$	$40 < x \leq 60$	$60 < x \leq 80$	$80 < x \leq 100$
frequency	4	20	31	34	11

- (a) Write down the modal class.

..... $< x \leq$ [1]

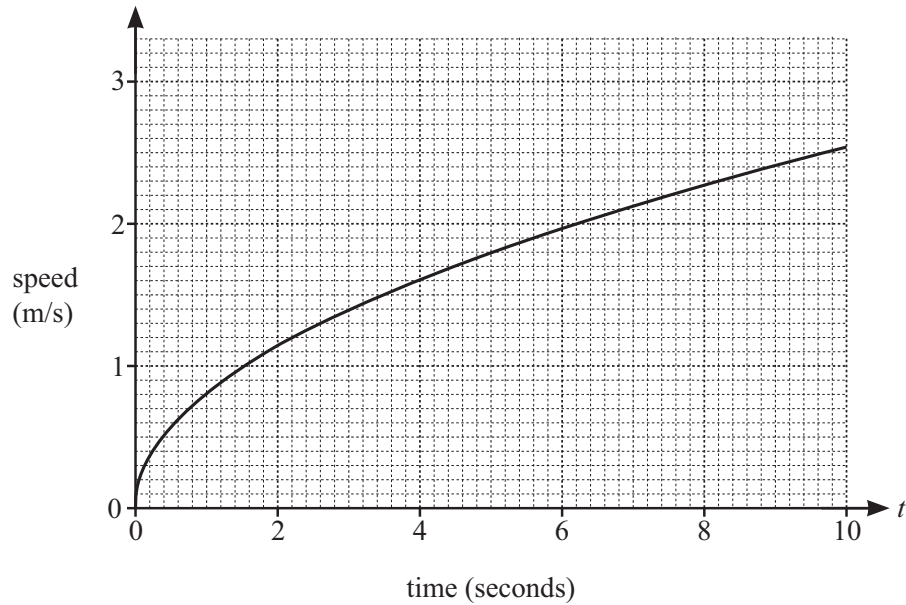
- (b) Calculate an estimate of the mean.

..... [4]



10 Llew cycles to school.

(a) The speed–time graph shows the first 10 seconds of his journey.



By drawing a suitable tangent, estimate the acceleration at $t = 2$.

..... m/s^2 [2]

(b) Llew cycles to school at an average speed of 18.9 kilometres per hour.

Write this speed in metres per second.

..... m/s [2]



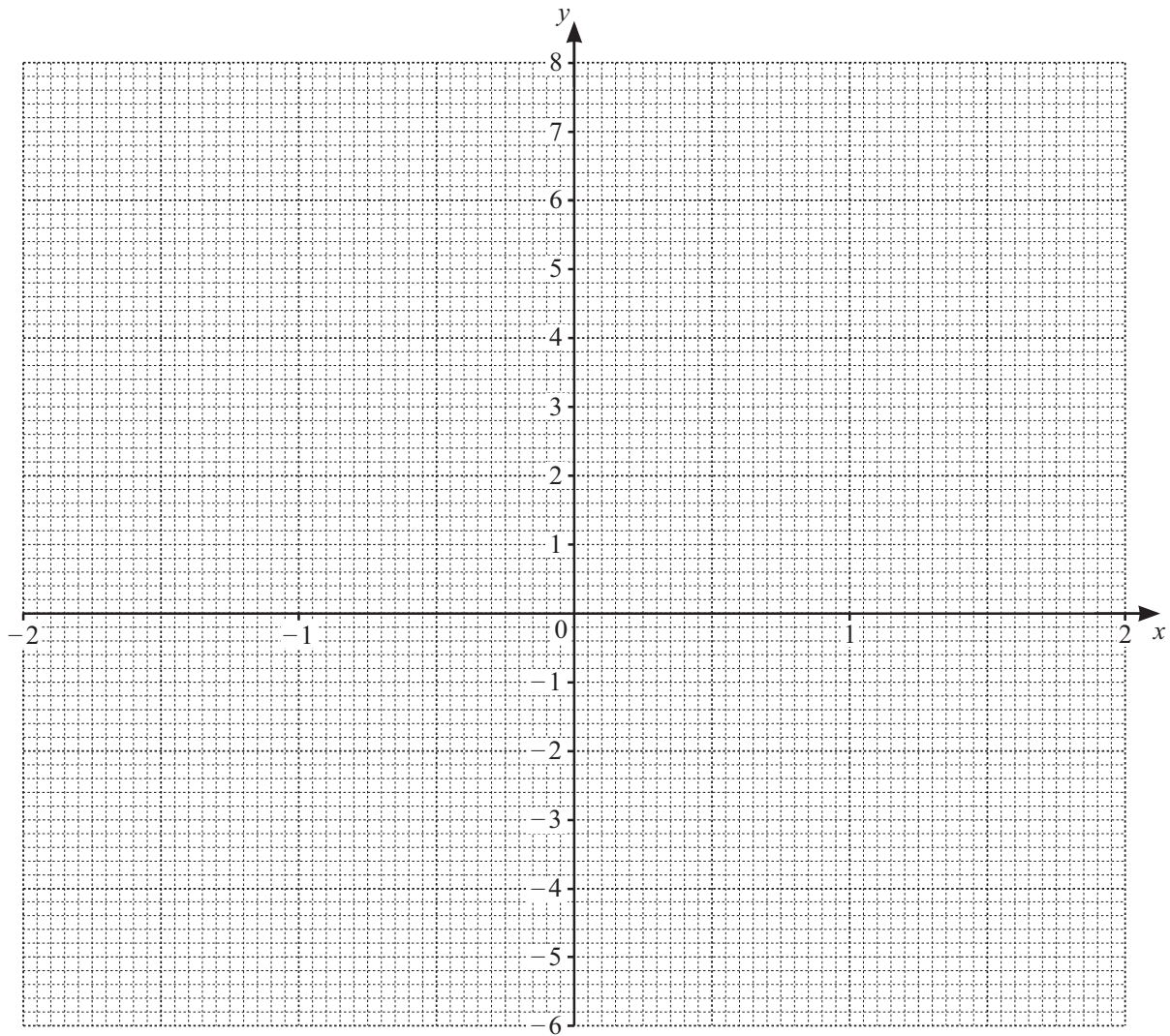
- 11 The table shows some values for $y = 3 \times 2^x - 5$ correct to 1 decimal place.

x	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2
y		-3.9		-2.9		-0.8	1	3.5	7

(a) Complete the table.

[3]

(b) Draw the graph of $y = 3 \times 2^x - 5$ for $-2 \leq x \leq 2$.



[4]





(c) By drawing a suitable line on the graph, solve the equation $3 \times 2^x = 7$.

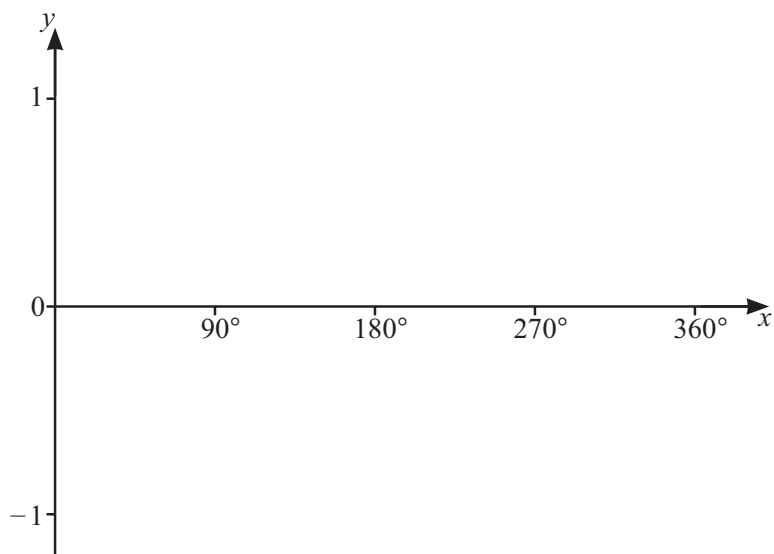
$x = \dots\dots\dots$ [2]

(d) Find the equation of the asymptote to the graph of $y = 3 \times 2^x - 5$.

$\dots\dots\dots$ [1]



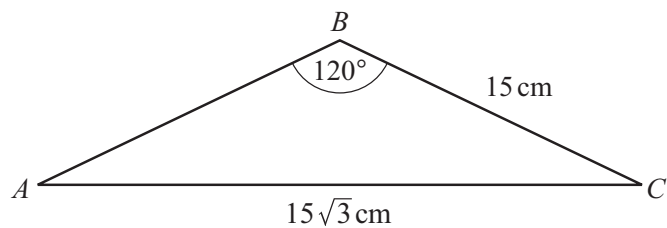
- 12 (a) Sketch the graph of $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$.



[2]

- (b) Solve the equation $3\cos x + 1 = 0$ for $0^\circ \leq x \leq 360^\circ$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [3]



NOT TO SCALE

The diagram shows triangle ABC .

$BC = 15$ cm, $AC = 15\sqrt{3}$ cm and angle $ABC = 120^\circ$.

(a) Show that triangle ABC is an isosceles triangle.

[4]

(b) Calculate the shortest distance from B to AC .

..... cm [3]





14 A is the point $(5, 2)$ and B is the point $(-3, 6)$.

(a) Calculate the length of AB .

..... [3]

(b) Find the equation of the line perpendicular to AB that passes through the point $(-2, 5)$.
Give your answer in the form $y = mx + c$.

$y =$ [4]





15

- 15 The sides of a triangle are 5 cm, 7 cm and 11 cm.

Find the size of the largest angle in this triangle.

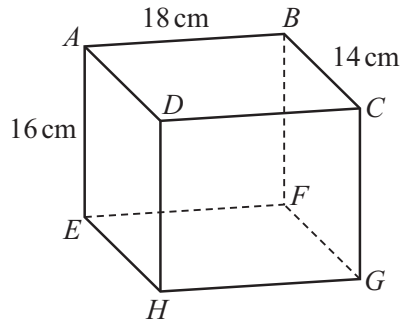
..... [3]

- 16 A rectangle measures 4.2 cm by 6.8 cm.
Each measurement is given correct to the nearest millimetre.

Calculate the lower bound of the area of the rectangle.

..... cm² [2]





NOT TO SCALE

The diagram shows a cuboid, $ABCDEFGH$.
 $AB = 18$ cm, $BC = 14$ cm and $AE = 16$ cm.

Calculate the length of the diagonal AG .

..... cm [3]





18 (a) $f(x) = 3x + 1$ $g(x) = 5 - 3x$

(i) Find x when $f(x) = g(x)$.

$$x = \dots\dots\dots [2]$$

(ii) Find $g^{-1}(x)$.

$$g^{-1}(x) = \dots\dots\dots [2]$$

(iii) Find $gg(x)$.
Give your answer in its simplest form.

$$gg(x) = \dots\dots\dots [2]$$

(b) $h(x) = 2^{2x}$

Find x when $h^{-1}(x) = 2$.

$$x = \dots\dots\dots [2]$$



- 19 (a) Lee invests \$2000 in an account that pays simple interest at a rate of 3.5% per year.

Work out the **total** value of the investment at the end of 4 years.

\$ [3]

- (b) Mia invests \$3000 in an account that pays compound interest at a rate of 3.2% per year.

Find the number of complete years it takes for the value of Mia's investment to become greater than \$4800.

..... [3]

- (c) Ryan invests some money.

The value of the investment increases exponentially at a rate of $r\%$ per year.

At the end of 6 years, the value of the original investment has increased by 19.4%.

Find the value of r .

$r =$ [3]



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DFD



- 20 (a) Find the coordinates of the turning points of the graph $y = 8x^3 - 6x + 5$.
You must show all your working.

(..... ,)

(..... ,)

[5]

- (b) Determine whether each turning point is a maximum or a minimum.
Show how you decide.

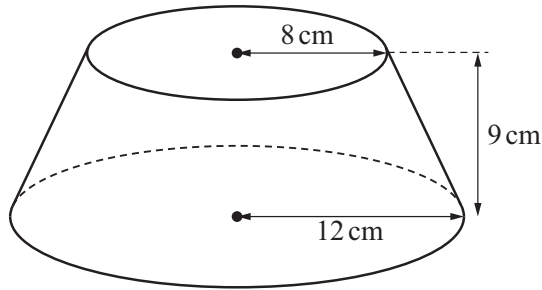
[3]

Question 21 is on the back page.





NOT TO SCALE



The diagram shows a frustum.

Calculate the volume of the frustum.

..... cm^3 [5]

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